

LDA TOPIC MODELING

COVID-19 TWEETS (from FL)

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Workflow



Preparing the data

- tidyverse, tidytext, qdap, tm, SnowballC, topicmodels
- Cleaning the tweets, tokenizing the words, stemming + stem completion



tidy(matrix = "beta")

- Trial-and-error'd several different k values, proceeding with the model which gave the most humanly interpretable results



tidy(matrix = "gamma")

- Determined the most overall talked about topics during April - June in 2020 and 2021 by assigning topics based off highest gamma per document

Preparing The Data (PT.1)

Original tweet:

@ayemojubar We have lost 3 healthcare workers: Dr Emeka, Dr(PT) Otegbeye and Dr Aminu...yet some still think COVID-... <https://t.co/u04st9avXw>

Semi-cleaned tweet:

We have lost three healthcare workers Dr Emeka DrPT Otegbeye and Dr Aminuyet some still think COVID...

Then:

- Tokenizing using **unnest_tokens()**
- Removing stop words using **anti_join()**

Preparing The Data (PT.2)

1) Word Stemming & Stem Completion

- A. Created a dictionary, pulling distinct words from the cleaned, tokenized dataframe
- B. `wordStem()` - breaking the words into their roots
- C. `stemCompletion()` - using the dictionary as reference to complete the words

Some words did not stem-complete the way I hoped, so I manually changed them using `mutate()` and `recode()`.

2) Counting:

`count(id, word)`

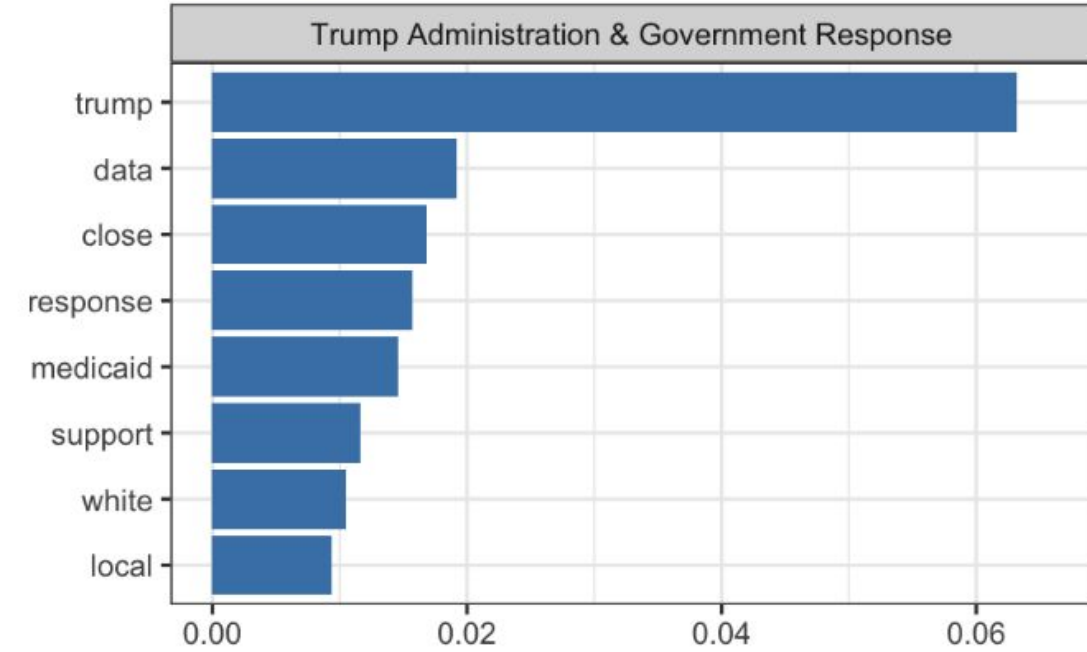
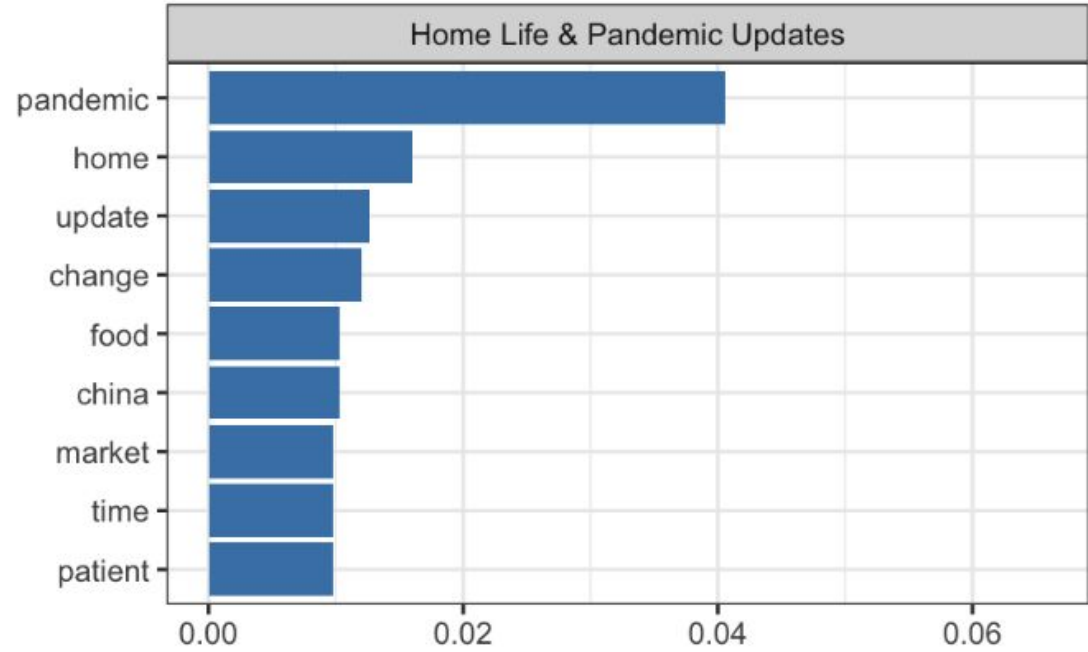
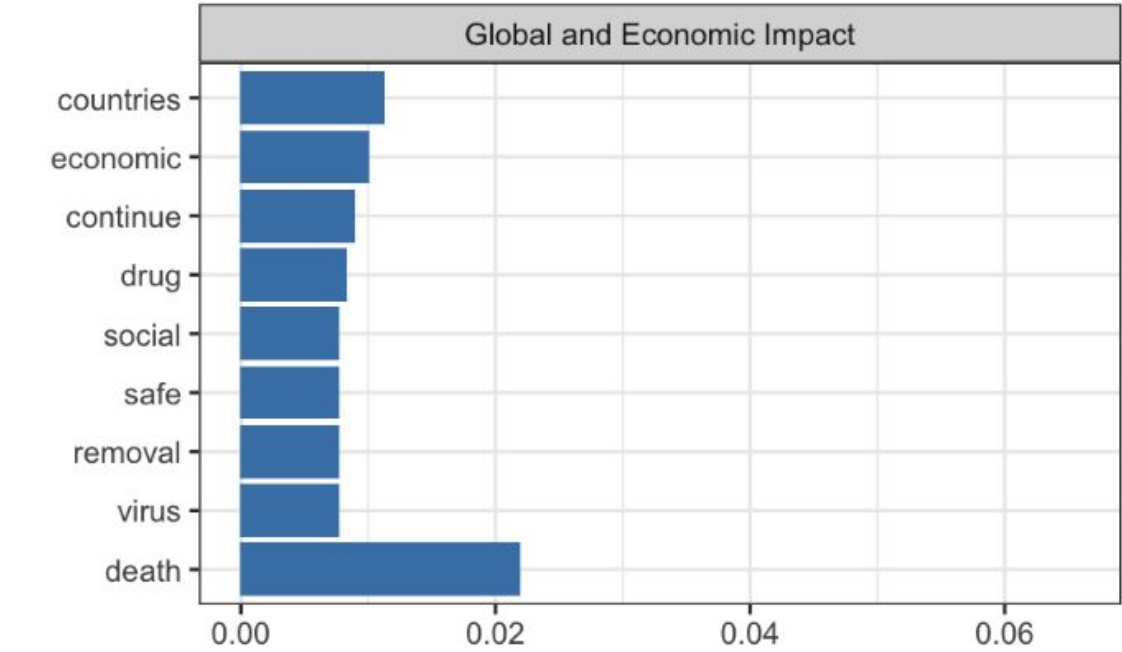
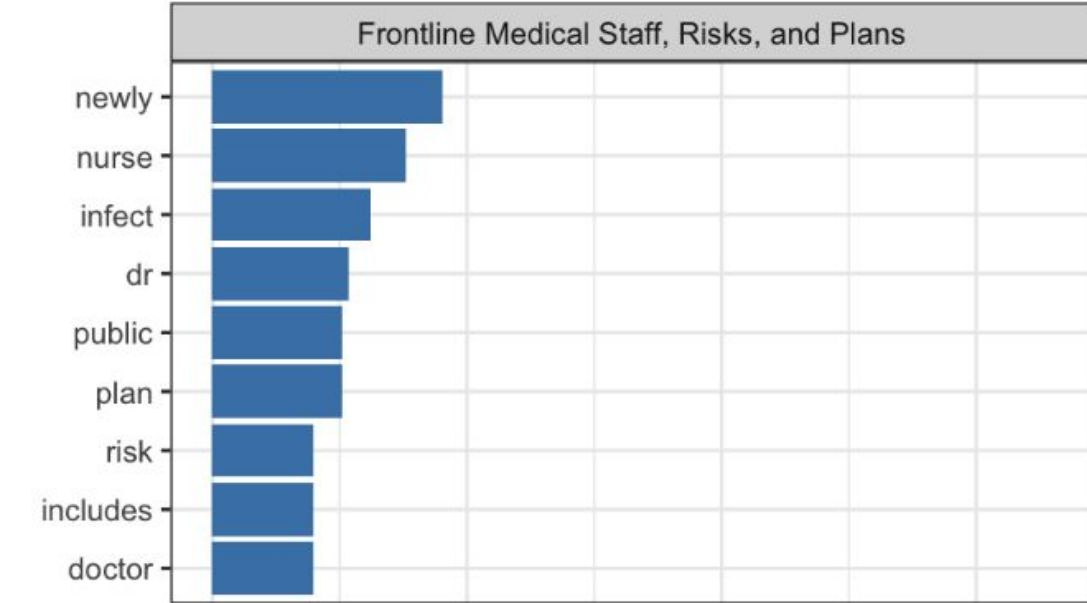
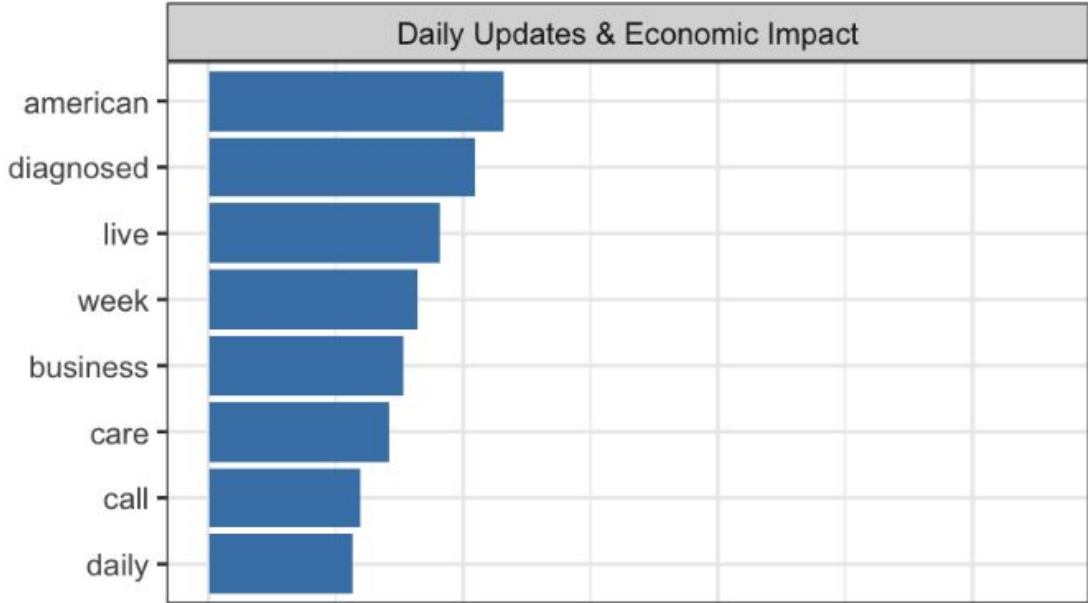
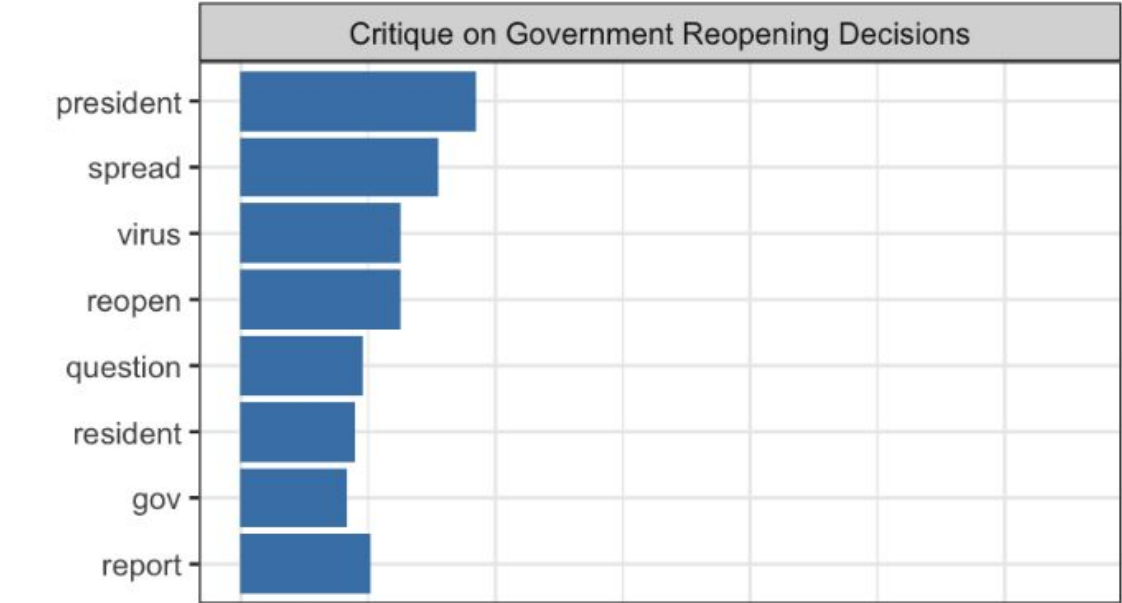
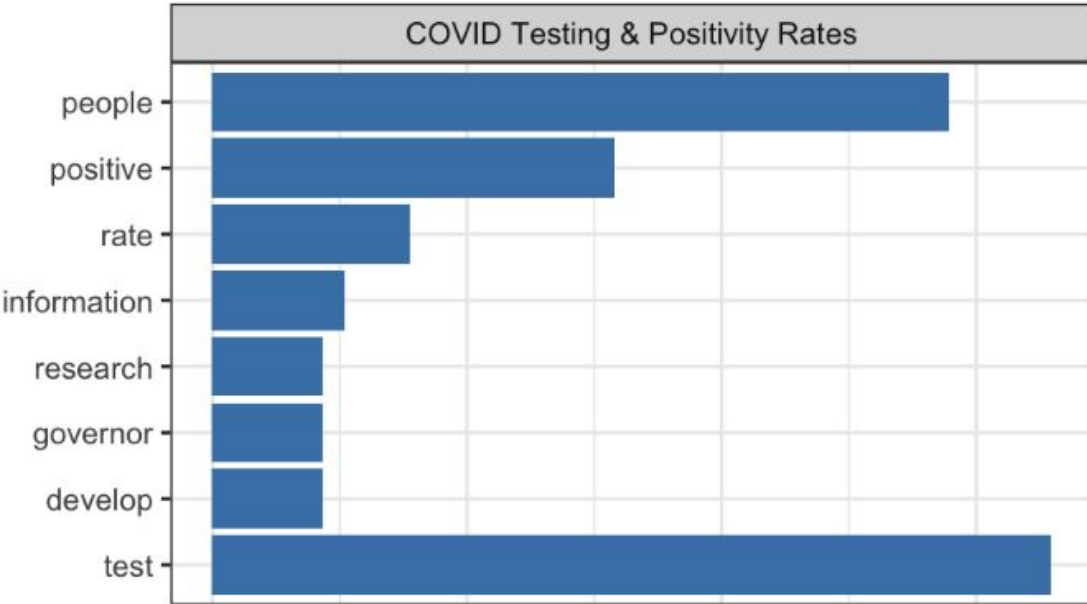
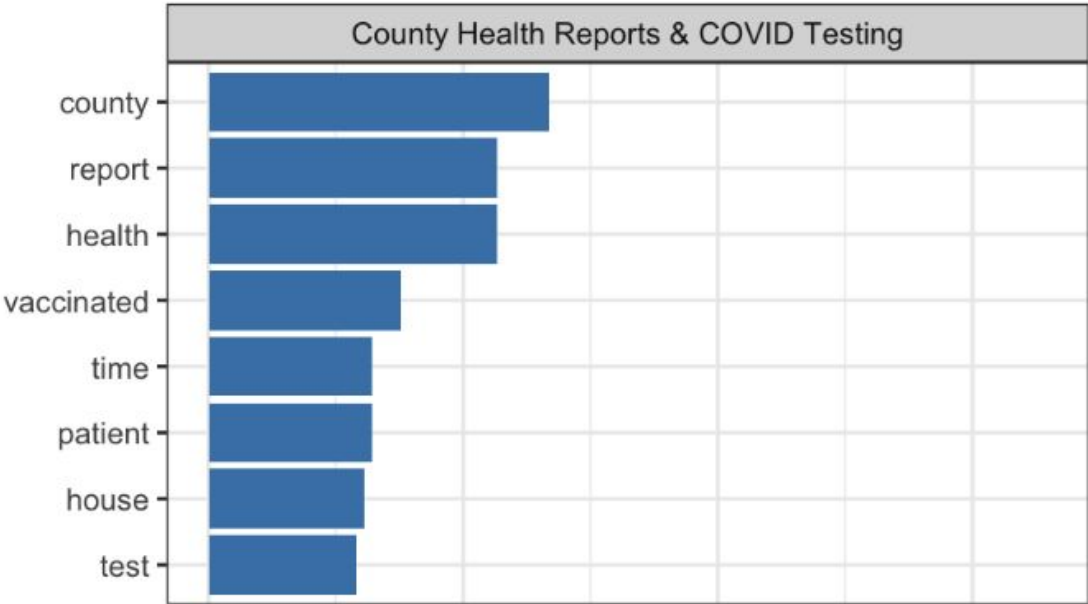
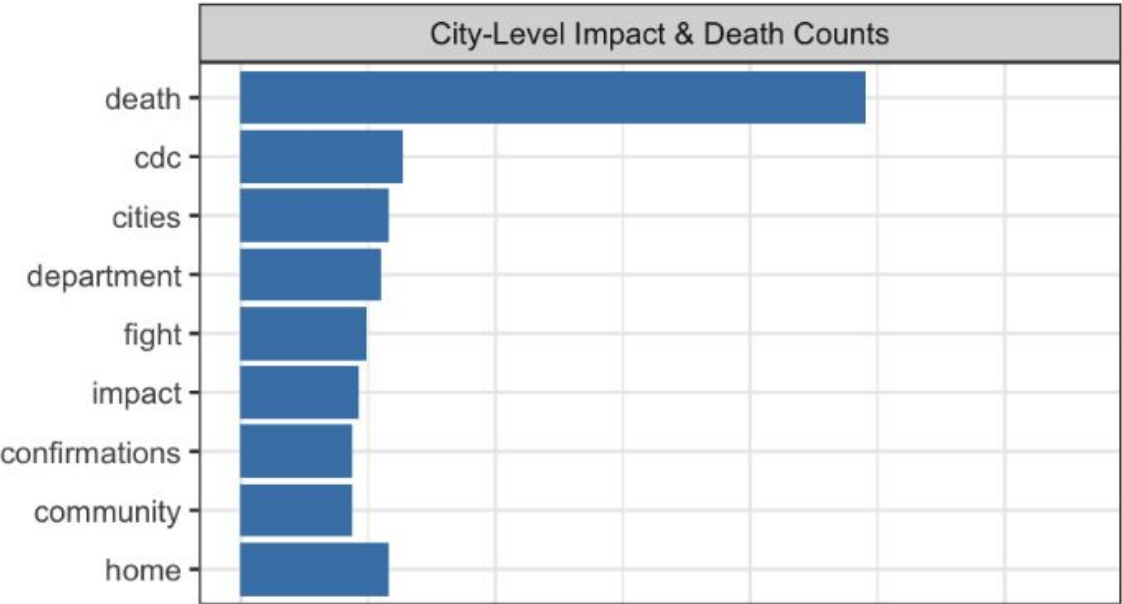
3) Creating DTM:

`cast_dtm(id, word, n)`

Latent Dirichlet Allocation

An unsupervised machine learning topic modeling technique.
It intends to uncover topics and themes which exist in and characterize the documents / text data.

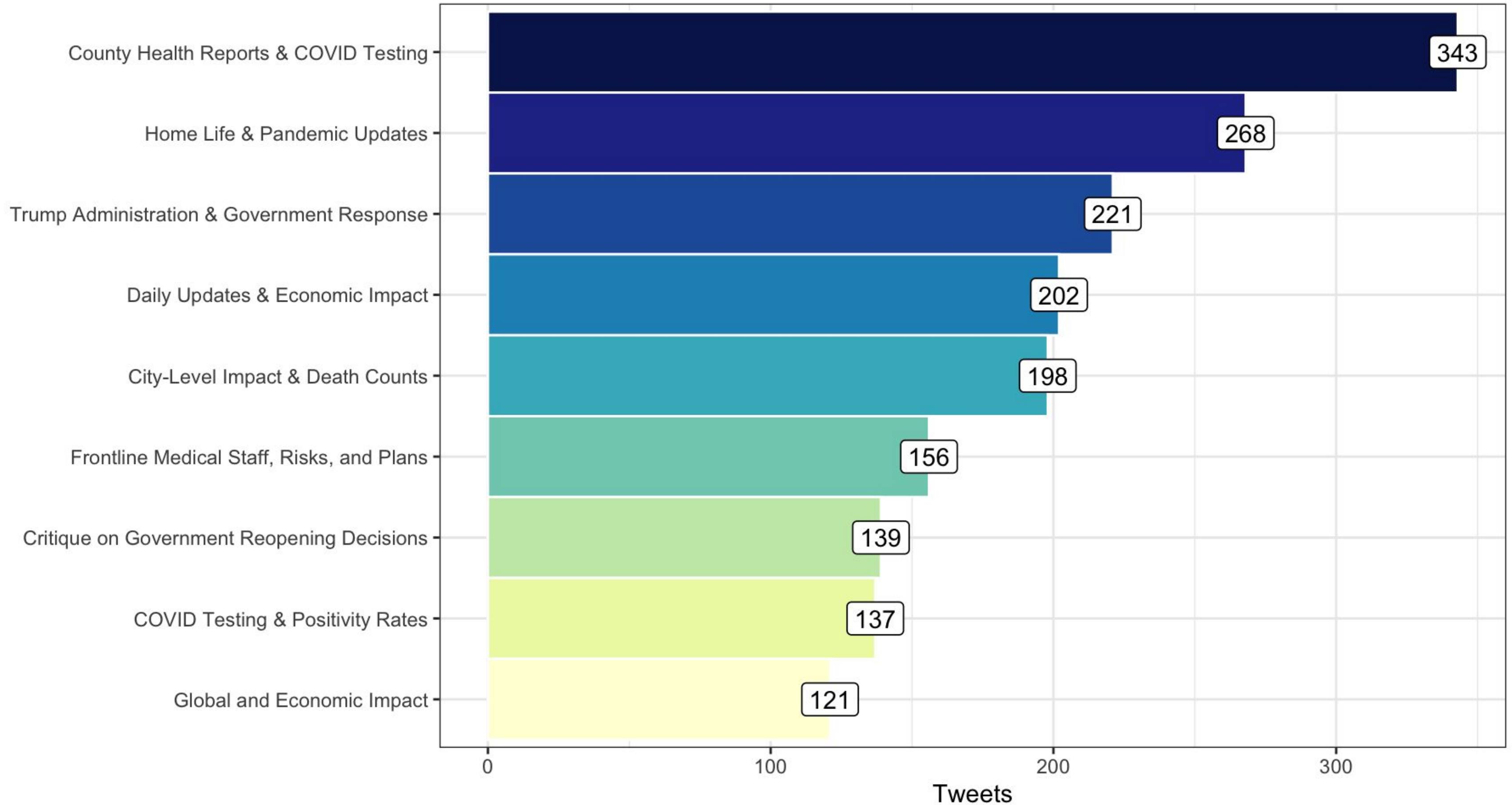
April - June 2020 Topics



Beta

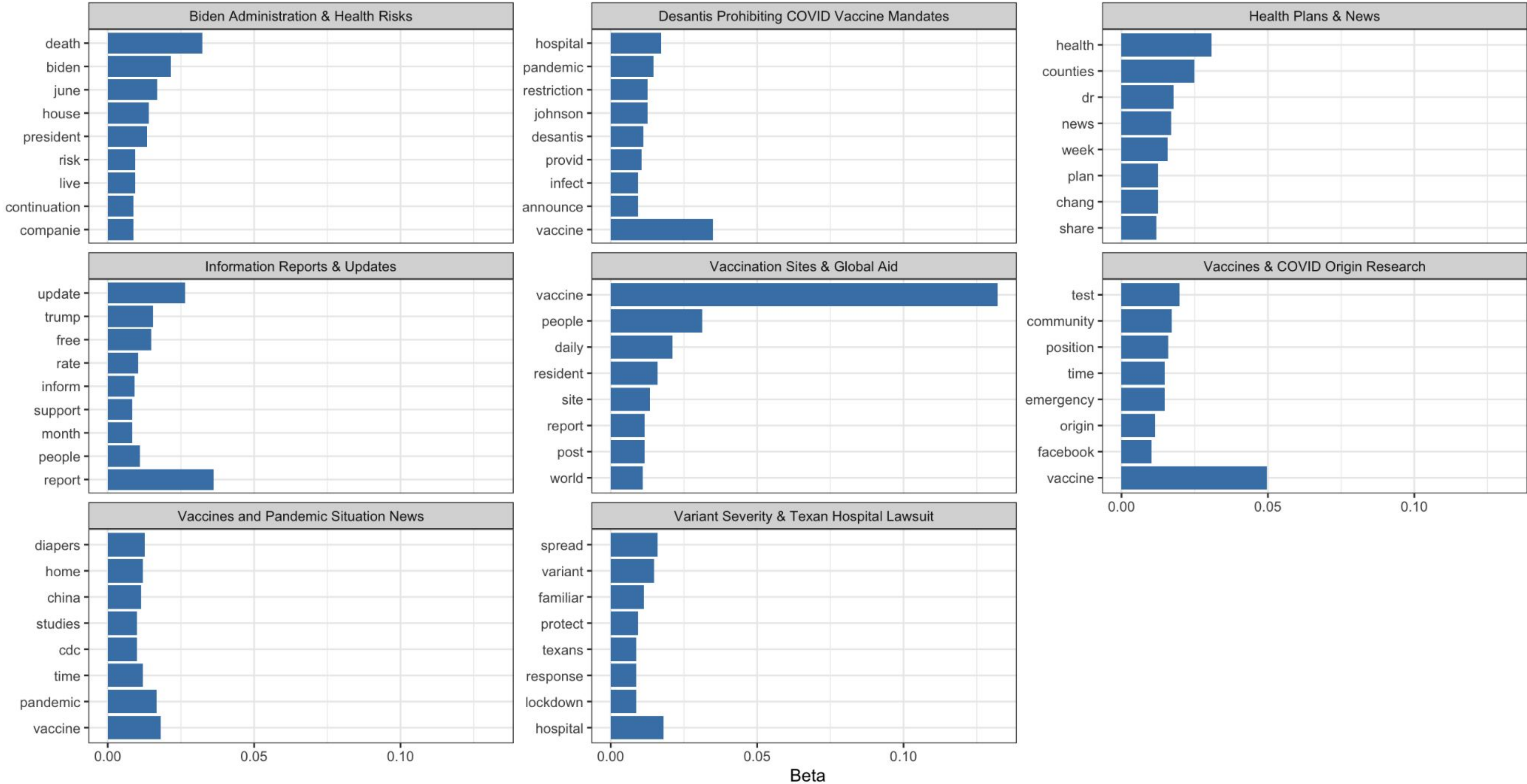
The Biggest Themes of April - June 2020

Relating to Covid-19



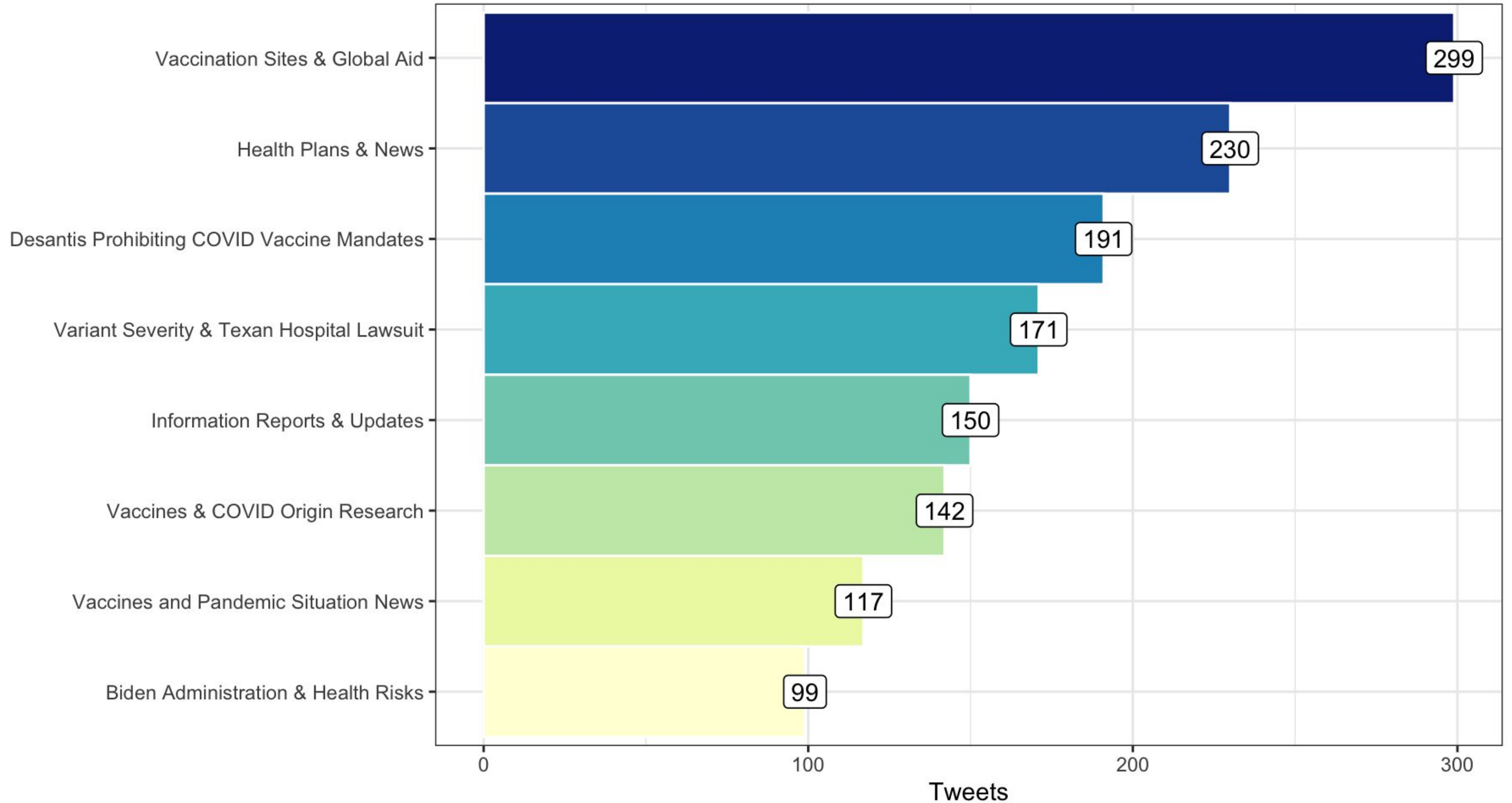
- For context: Early April, the CDC announced new mask-wearing guidelines. Many people complied; current President Trump did not. Florida ended quarantine on April 30th and moved into phases of reopening the public. In early June, the state had entered phase two of reopening, meaning allowing bars, theaters, and other public building spaces to run at 50% capacity. There was no vaccine yet, but development, research, and human clinical trials had been ongoing and fast.
- The most prominent topics that Floridians, likely like most people, talked about during April-June of 2020 are:
 - County Health Reports & COVID Testing - Because the global pandemic is new to most people, and many people across the globe were experiencing severe illness and death counts kept rising, there were so many updates and some many people watching the numbers climb. I remember back in my senior of high school where I was checking covid cases every day. With just one case in your state, it spreads like wildfire—both the virus and news.
 - Home Life and Pandemic Updates - Amidst the reopening and returning to a more normal life again, it seems many conversations and tweet documents showed people's reliance on official pandemic updates to decide how to move forward. Reopening with no mask mandates in Florida, there was a surge in cases. Keep in mind, again, that there was no vaccine, which definitely led to more patients. The word 'China' may have to do with the origin of the virus or blaming the country.
 - Trump Administration & Government Response - We were under Trump's presidency during the time these tweets were collected. During this period, the White House held frequent briefings and released evolving public health guidelines. Additionally, during March, Medicaid temporarily expanded tele-health benefits. This provided a wider range of healthcare to the beneficiaries with the need to travel to a healthcare facility.
- More can be found here: <https://www.cdc.gov/museum/timeline/covid19.html>
- Overall, many of the tweets consisted of reporting official news, health guidelines and also a lot of conversations around these new experiences, such as the entire pandemic and home life. Reopening after quarantine, some people were pre-cautious and others were tired of quarantine and masks.

April - June 2021 Topics



The Biggest Themes of April - June 2021

Relating to Covid-19



- In contrast to the previous topics found in tweets from mid-2020, the topics of 2021 is dominated by the existence of vaccines. That makes sense. There was no covid vaccine while investigating the tweets from 2020, since the first covid vaccine came out in the December of 2020.
- Some context: At this time, we were under the Biden administration, which joins COVAX—program aimed at vaccinating people in low and middle income countries against COVID-19. Hence the most prevalent topic in 2021 regarding covid: Vaccination Sites & Global Aid. Vaccines were being distributed more, nationally and internationally. Also, by spring of 2021, about most age groups were able vaccinated, including senior citizens, adults, and adolescents.
- There is strong pushback for vaccines in general, but during 2021, a variant like delta came to be. It was more contagious and more severe, but a large population of people still refused to wear masks and get vaccinated, fearing the long term complications due to limited testing. Desantis officially prohibited vaccine and mask mandates.
- Ultimately, comparing the tweets and their topics with a one year gap highlights vaccine developments. In the 2020 tweets, Florida was just starting to reopen. And in 2021, life felt more normal, at least for the U.S. (hence less conversations on the national death reports.) The same cannot be said about lower income countries globally, which relied of other countries' distribution of vaccines. Despite vaccines being a game changer, allowing the reopening of society to function so well, masks were and vaccines still are very controversial. The theme of vaccines heavily dominated several topics in 2021.

SUMMARY INSIGHTS

For 2020, the strongest themes regarded health and test reports, pandemic updates, and government response. However, for 2021, we can observe the theme of vaccines dominating most topics.